Education for Chemical Engineers xxx (xxxx) xxx

Contents lists available at ScienceDirect

Education for Chemical Engineers

journal homepage: www.elsevier.com/locate/ece

Critique: YEASTsim - A Matlab-based simulator for teaching process control in fed-batch yeast fermentations

Juan Sebastian Jaramillo, Luis H. Reyes *

Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

A R T I C L E I N F O

Keywords:
MATLAB-based simulator
Process control
Fed-batch fermentation
Chemical
Reaction engineering
Saccharomyces cerevisiae
Simulation-based learning
Educational tools
Bioprocess modeling
Chemical engineering education

A B S T R A C T

One of the primary challenges in chemical engineering education lies in the practical application of theoretical knowledge. Contemporary society demands chemical engineers who are not only adept at critical thinking but also proficient in practical problem-solving. Historically, acquiring such practical experience has been heavily dependent on laboratory experiments. However, with the advent of advanced technology, there are now more accessible and varied opportunities for experiential learning. Simulation-based learning tools have proven to be powerful complements to traditional laboratory experiments. These tools enable students to investigate complex chemical processes, test hypotheses, and develop practical skills in a safe, cost-effective, and scalable environment. This critique aims to evaluate the educational potential of the YEASTsim simulator, positioning it as an interactive and innovative tool to enhance the study of core subjects such as Chemical Reaction Engineering, a fundamental component of chemical engineering practice.

1. Generalities and GUI

YEASTsim is a MATLAB-based simulator developed by Dr. Hrnčiřík and Dr. Kohout from the Department of Mathematics, Informatics, and Cybernetics at the University of Chemistry and Technology Prague. The software is designed to model the fed-batch aerobic cultivation process of the baker's yeast Saccharomyces cerevisiae, a ubiquitous biotechnological process with significant applications in the food, pharmaceutical, and other industries (Hrnčiřík and Kohout, 2024). This initial version of YEASTsim encompasses a range of features and capabilities, which will be examined through several case studies to illustrate the simulator's potential and versatility.

Using well-known and simple growth models like Monod kinetics helps illustrate non-elementary reaction rates effectively. With default laboratory values, the simulator shows expected concentration profiles for substrate and biomass. However, the ethanol concentration profile shows an unusual oscillatory behavior, which is atypical for standard fermentation processes. By offering a user-friendly graphical user interface (GUI), YEASTsim allows users to manipulate variables and view results seamlessly, making it a valuable educational tool. The GUI's design ensures ease of use, which is crucial for effective learning and teaching in chemical engineering. The inclusion of clear, detailed graphs and well-labeled axes enhances the user experience, providing students with a practical, hands-on approach to understanding complex chemical processes (Komulainen et al., 2012).

In an initial examination of the YEASTsim simulator, its exceptional user-friendliness stands out, largely due to its graphical user interface (GUI). This interface allows users to effortlessly manipulate variables, which is essential for educational applications. Designed to be intuitive, the GUI ensures that users can easily navigate and engage with the simulator. Furthermore, it provides clear and detailed graphs of output variables, giving users the option to select which graphs to display. The use of vibrant colors and well-labeled axes enhances the distinction and clarity of the information presented. This combination of features renders the simulator a highly suitable tool for teaching and learning.

However, it would be highly advantageous if users could control the manipulated variables—such as feed, air flow rate, and stirrer speed—directly from the YEASTsim interface. Dissolved oxygen, a critical parameter in control strategies, is typically regulated by these variables, which significantly influence the yield and productivity of the reaction (Gao and Lin, 2016). While these variables can be modified directly within the MATLAB code, this method lacks intuitive usability. Therefore, enabling users to adjust these variables through the interface would streamline the process, facilitating easier analysis of their impact on the output variables and enhancing the overall user experience.

* Corresponding author.
E-mail address: lh.reyes@uniandes.edu.co (L.H. Reyes).

https://doi.org/10.1016/j.ece.2024.06.005
Received 28 June 2024; Accepted 30 June 2024

2. Cases of study

2.1. Variety of control strategies

YEASTsim 1.0 offers multiple control strategies for the fermentation process, encompassing both feedback (rule-based control strategy using dissolved oxygen concentration, DOCONT) and feedforward control strategies (Hrnčiřík and Kohout, 2024). Users can study these strategies interchangeably to understand how profiles, yield, and productivity vary with different control methods. A compelling area for investigation is the sensitivity of each control strategy. Sensitivity, in this context, refers to the responsiveness of output variables to specific input parameters, such as fermentation time, sampling period, glucose concentration, and manipulated variables under different control strategies. Grasping sensitivity is crucial for comprehending not only the influence of these parameters on process outcomes but also their interactions and potential impacts on system stability and performance. This exploration can yield deeper insights into optimizing fermentation processes and developing more robust control strategies.

For those interested in implementing novel and diverse control strategies, YEASTsim provides a significant advantage by including code examples and templates for creating user-defined control strategies and embedding them into the simulator via MATLAB scripts. These resources offer a foundational platform for users to customize the simulator to their specific needs, such as examining sensitivity. For example, investigating the control strategies proposed by Hisbullah et al. (2001), which utilize a hybrid neural-network PI controller, may offer valuable insights and help mitigate potential oscillations or offsets in the profiles. Such explorations enhance a student's understanding of control theory and its applications in chemical reaction engineering, particularly in fermentation processes.

2.2. Kinetics

When examining the kinetics of the fermentation process, it is noteworthy that YEASTsim incorporates the three metabolic pathways of *Saccharomyces cerevisiae*: oxidative growth on glucose, reductive growth on glucose, and oxidative growth on ethanol (Sonnleitner and Käppeli, 1986). Utilizing well-established and straightforward growth models such as Monod kinetics effectively illustrates non-elementary reaction rates. With default laboratory values, the simulator generates expected concentration profiles for substrate and biomass. However, the ethanol concentration profile exhibits an unusual oscillatory behavior (see Fig. 1), which is atypical for standard fermentation processes.

In a well-controlled fermentation, the ethanol concentration typically increases steadily as the yeast converts glucose into ethanol. This irregularity in the ethanol profile necessitates thorough investigation, as it is observed exclusively with DOCONT strategies v1, v2, v3, and v4, whereas the feedforward strategy yields the expected behavior. Although the primary focus of the software is on biomass growth, it is imperative not to overlook ethanol production. This irregular ethanol profile suggests an issue that must be addressed to ensure the accuracy and reliability of the simulator.

YEASTsim effectively demonstrates the impact of parameters such as glucose concentration in the substrate feed rate on a fed-batch fermentation process, highlighting its significant influence on metabolic rates and volume dynamics. According to Yüzgeç (2010), if the glucose concentration in the fermentor is excessively high, the final product concentration will be elevated; however, the yield will be low due to excessive ethanol formation. Conversely, if the glucose concentration is insufficient, production will be low, but the yield will be high. This feature underscores the simulator's effectiveness in illustrating critical biochemical principles for educational purposes.

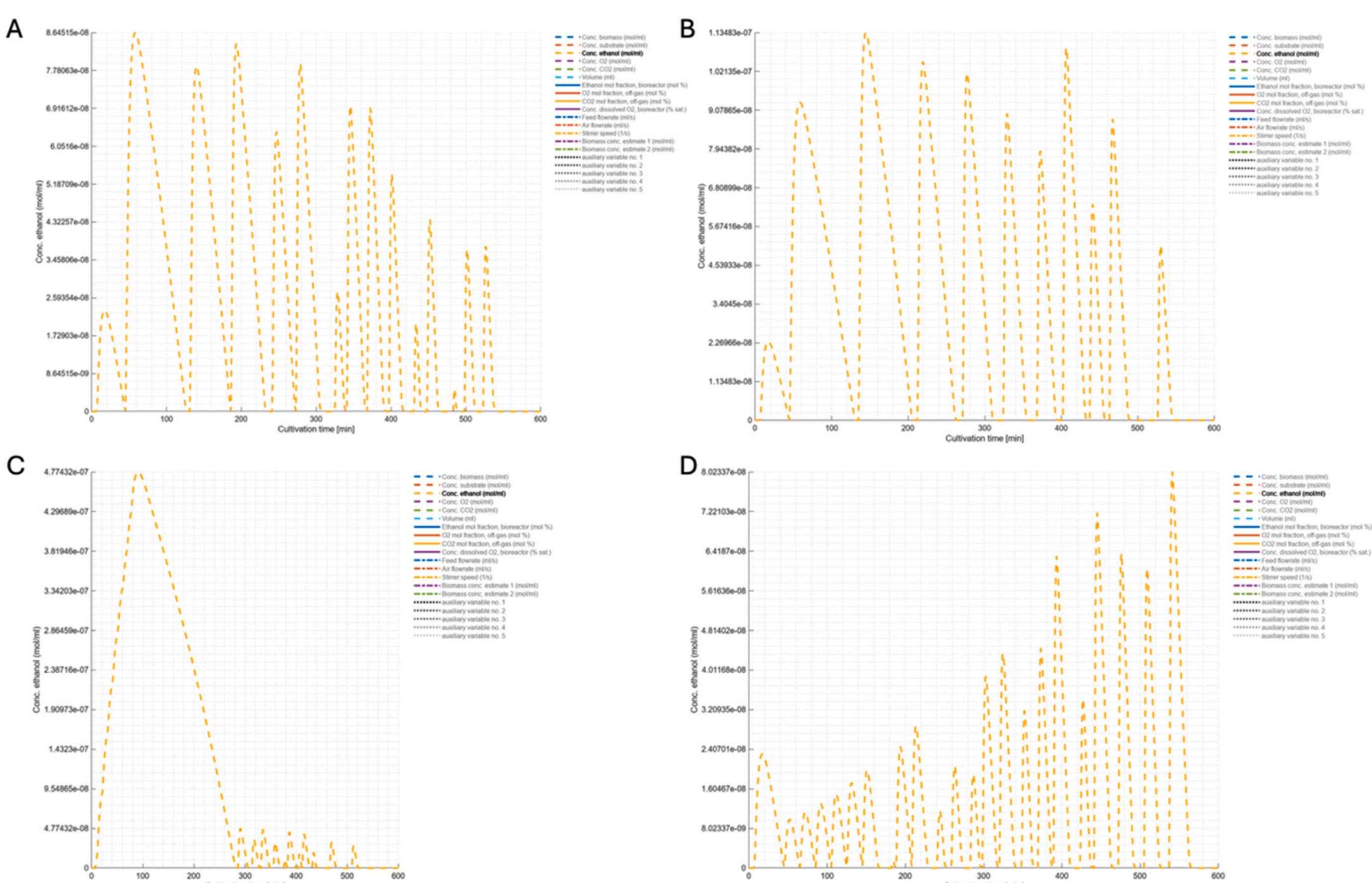

Fig. 1. Ethanol concentration profiles displayed by YEASTsim with default values for each DOCONT strategy: a) v1, b) v2, c) v3, and d) v4.

2.3. Design parameters and cultivation time

One of the primary objectives of a chemical reaction engineering course is to design and size reactors based on appropriate mathematical models for their operation. For instance, parameters such as the volumetric mass transfer coefficient, typically defined using empirical correlations, are known to function effectively only within a specific range of fermentor sizes (Fuchs et al., 1971). The authors reference a 7.5-liter bioreactor as their benchmark; however, to enhance the educational utility of YEASTsim, it is recommended that the guide includes the acceptable range of fermentor volumes at bench scale. This addition would be particularly valuable when encountering unusual output values or concentration profiles.

Moreover, cultivation time is another critical variable that can be explored. When simulating extended periods, the bioreactor often reaches its maximum capacity due to the control strategies, causing the output graphs to display unusual values. Analyzing the effect of residence time on yield and productivity helps determine the optimal moment for the reactor to switch to a fully batch mode for safety reasons. This approach would allow users to understand how reactor volume affects the output, aiding students in visualizing and interpreting these effects within the given parameters.

3. Other software limitations and recommended future approaches

Regarding scalability, the YEASTsim simulator can be adapted to model fermentation reactors at various scales. Users have the flexibility to modify the MATLAB code, enabling future versions of the software to incorporate different equations and correlations based on the volume of the bioreactor, thus extending its applicability beyond just lab-scale scenarios. By adjusting model parameters and control strategies, the simulator can eventually be utilized to study conditions for industrial-scale fermentation processes, offering a valuable resource for both educational and research purposes.

The YEASTsim simulator, in its version 1.0, provides a dynamic, robust, and practical tool for simulating fermentation processes. However, the current model may have limitations that prevent it from accurately capturing real-world behavior. Specifically, the software omits a crucial variable: temperature. The current documentation and MATLAB code do not specify the temperature conditions under which the bioreactor operates. The exothermic nature of the reaction is not accounted for due to the absence of an energy balance within the system of ordinary differential equations. Temperature is a critical factor in the growth of baker's yeast. According to Lin et al. (2012), experimental data on Saccharomyces cerevisiae alcoholic fermentation showed that excessively high temperatures inhibited cell growth, leading to a significant decline in fermentation efficiency. Specifically, at 45°C, the system exhibited high rates of cell growth and ethanol production, but these rates were inhibited at 50°C. Optimal specific growth rates and ethanol production were observed between 30°C and 45°C, depending on the initial glucose concentrations. Not maintaining these conditions could result in undesirable outcomes. Allowing users to experiment with different initial temperatures and incorporating temperature control strategies to keep the bioreactor within ideal temperature ranges would be a valuable enhancement to the simulator.

Applying a fermentor simulation like YEASTsim can significantly enhance the educational experience by providing software that allows students to engage with complex chemical reaction engineering concepts. YEASTsim provides an interactive environment for simulating fed-batch fermentation processes, empowering students to explore complex phenomena, test hypotheses, and conduct in-depth analyses and manipulations of fermentation processes. Identified areas for improvement in future versions include incorporating temperature control features, direct manipulation of critical process variables, and more explicit guidance on acceptable bioreactor volume ranges. Overall, YEASTsim 1.0 represents a commendable advancement in simulation-based learning, demonstrating its potential to bridge the gap between theoretical knowledge and practical proficiency in chemical engineering education.

CRediT authorship contribution statement

Luis H. Reyes: Conceptualization, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. **Juan Sebastian Jaramillo:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest regarding the publication of this manuscript.

Acknowledgments

The authors express their gratitude to the Department of Chemical and Food Engineering at Universidad de los Andes for their invaluable support and resources. The authors also acknowledge the utilization of ChatGPT-4.0 as a grammar checker and style enhancer. ChatGPT-4.0's advanced language processing capabilities were instrumental in refining the manuscript, ensuring clarity, coherence, and a formal academic tone appropriate for publication.

References

Fuchs, R., Ryu, D.D.Y., Humphrey, A.E., 1971. Effect of surface aeration on scale-up procedures for fermentation processes. Ind. Eng. Chem. Process Des. Dev. 10, 190–196. https://doi.org/10.1021/i260038a009.

Gao, L., Lin, H., 2016. Simulation of feedforward-feedback control of Dissolved Oxygen (DO) of microbial repeated fed-batch culture, 4.1-4.6 Int. J. Simul. Syst., Sci. Technol. 17. https://doi.org/10.5013/IJSSST.a.17.14.04.

Hisbullah, M.A., Hussain, M., Ramachandran, K.B., 2001. Comparative evaluation of various control schemes for fed-batch fermentation. Bioprocess Biosyst. Eng. 24, 309–318. https://doi.org/10.1007/s00449-001-272-7.

Hrnčiřík, P., Kohout, J., 2024. YEASTsim: a Matlab-Based Simulator for the Study of Process Control of Fed-batch Yeast Fermentations. Department of Mathematics, Informatics and Cybernetics, University of Chemistry and Technology Prague.

Komulainen, T.M., Enemark-Rasmussen, R., Sin, G., Fletcher, J.P., Cameron, D., 2012. Experiences on dynamic simulation software in chemical engineering education. Educ. Chem. Eng. 7, e153–e162. https://doi.org/10.1016/j.ece.2012.07.003.

Lin, Y., Zhang, W., Li, C., Sakakibara, K., Tanaka, S., Kong, H., 2012. Factors affecting ethanol fermentation using Saccharomyces cerevisiae BY4742. Biomass Bioenergy 47, 395–401. https://doi.org/10.1016/j.biombioe.2012.09.019.

Sonnleitner, B., Käppeli, O., 1986. Growth of Saccharomyces cerevisiae is controlled by its limited respiratory capacity: formulation and verification of a hypothesis. Biotechnol. Bioeng. 28, 927–937. https://doi.org/10.1002/BIT.260280620.

Yüzgeç, U., 2010. Performance comparison of differential evolution techniques on optimization of feeding profile for an industrial scale baker's yeast fermentation process. ISA Trans. 49, 167–176. https://doi.org/10.1016/j.isatra.2009.10.006.